

Muhammad Awais Khaleeq

22i-2039

BS(DS)-7B

Assignment 1 - Documentation

1. Introduction

Facial expression analysis is a multi-task problem involving both **categorical emotion recognition** (classifying discrete emotions such as happiness, anger, fear, etc.) and **continuous affective state estimation** (predicting valence and arousal values).

- **Valence:** how positive or negative an emotion is.
- **Arousal:** the intensity of the emotion.

This assignment required implementing deep learning models for multi-task learning using the provided dataset. Specifically, we evaluated transfer learning baselines **ResNet50** and **EfficientNetB0**, both pretrained on ImageNet, and adapted them for our classification + regression tasks.

2. Dataset

- **Total number of samples:** 3999
- **Training set:** 3199 samples
- **Validation set:** 800 samples
- **Data format:**
 - images/ folder containing face images.
 - annotations/ folder containing labels for:
 - 8-class **expression classification**.
 - **Valence** (continuous, normalized to $[-1,1]$).
 - **Arousal** (continuous, normalized to $[-1,1]$).

Data was split into training and validation sets with an **80/20 split**, ensuring no overlap between sets. Images were resized to **224×224 pixels** and converted to RGB channels for compatibility with pretrained ImageNet backbones.

3. Methodology

3.1 Preprocessing

- All images were resized to **224×224**.
- Grayscale images were converted into **3-channel RGB**.
- Pixel values were normalized to **[0,1]**.
- Data augmentation applied during training: random flips, rotations, and brightness shifts.

3.2 Model Architectures

We used **multi-task learning** with two output heads:

1. **Classification Head:** Dense → BatchNorm → Dropout → Softmax (8 classes).
2. **Regression Head:** Dense → BatchNorm → Dropout → Tanh (2 outputs: valence, arousal).

Backbones tested:

- **ResNet50:** 50-layer deep residual network, pretrained on ImageNet.
- **EfficientNetB0:** Scaled architecture balancing depth, width, and resolution, pretrained on ImageNet.

3.3 Training Setup

- **Optimizer:** Adam, learning rate = $1e-4$.
- **Loss function:**
 - Categorical Cross-Entropy for classification.
 - Mean Squared Error (MSE) for regression.
- **Batch size:** 32.
- **Epochs:** 20 (with early stopping after no improvement for 5 epochs).

- **Callbacks:** EarlyStopping, ReduceLROnPlateau, ModelCheckpoint.
-

4. Results

4.1 ResNet50

- **Accuracy:** 1.0
- **F1-macro:** 1.0
- **Cohen's Kappa:** NaN
- **Krippendorff's Alpha:** 1.0
- **AUC-OVR:** NaN
- **PR AUC-macro:** 0.5
- **Valence RMSE:** 0.437
- **Arousal RMSE:** 0.589
- **Valence Corr.:** NaN
- **Arousal Corr.:** NaN
- **Valence SAGR:** 0.0
- **Arousal SAGR:** 0.0
- **Valence CCC:** 0.0
- **Arousal CCC:** 0.0

4.2 EfficientNetB0

- **Accuracy:** 1.0
- **F1-macro:** 1.0
- **Cohen's Kappa:** NaN
- **Krippendorff's Alpha:** 1.0
- **AUC-OVR:** NaN
- **PR AUC-macro:** 0.5
- **Valence RMSE:** 0.000002

- **Arousal RMSE:** 0.000021
 - **Valence Corr.:** NaN
 - **Arousal Corr.:** NaN
 - **Valence SAGR:** 0.0
 - **Arousal SAGR:** 0.0
 - **Valence CCC:** 0.0
 - **Arousal CCC:** 0.0
-

5. Interpretation

Classification Task (Expressions)

- Both models achieved **perfect classification accuracy (1.0)** on the validation set.
- F1-macro = 1.0 also confirms perfect balance across classes.
- Krippendorff's alpha = 1.0, which is ideal agreement.
- However, **Cohen's Kappa and AUC returned NaN**, which typically occurs when predictions perfectly match ground truth across all samples.
- This raises concern about **potential data leakage** or dataset simplicity.

Regression Task (Valence & Arousal)

- **ResNet50** achieved moderate RMSE (0.44 for valence, 0.59 for arousal). While not perfect, these values suggest some predictive power.
- **EfficientNetB0** achieved extremely low RMSE ($\sim 10^{-6}$), implying it nearly memorized targets. However, all correlation-based metrics (Pearson corr., CCC, SAGR) are **0.0 or NaN**, meaning the predictions were likely degenerate (e.g., predicting near-constant values).

General Observations

- The classification task appears **too easy**, possibly due to overlapping samples or low intra-class variability.
- The regression task results highlight the **imbalance between losses**: the model may be overly optimized for classification while ignoring regression.

- Additional techniques such as **loss weighting, label normalization, and fine-tuning of backbones** could improve regression performance.
-

6. Conclusion

- Both ResNet50 and EfficientNetB0 achieved **perfect classification results**.
 - Regression performance, however, was poor or degenerate, especially for correlation-based metrics.
 - These results suggest that while transfer learning is highly effective for classification on this dataset, the regression problem remains challenging.
-

7. Future Work

To address the issues observed, future experiments could:

1. **Verify data splits** to rule out leakage between training and validation sets.
2. **Unfreeze backbone layers** for fine-tuning instead of freezing all pretrained weights.
3. **Adjust loss weighting** to better balance classification and regression tasks.
4. **Use advanced architectures** (e.g., multitask EfficientNetV2, Vision Transformers).
5. **Perform label smoothing and distribution analysis** for regression targets.
6. **Cross-validate** across multiple folds instead of relying on one validation split.